



MIDDLE CHILDHOOD

6–12 years

PowerPoint credit to: Rachael Pond

1

OVERVIEW

- What is middle childhood?
- Physical development
- Cognitive development
- Social-emotional development
- Peer relationships
- Bullying
- Protective & risk factors
- Children's perspectives of wellbeing

2

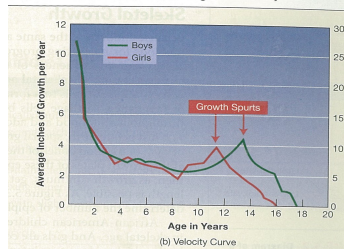
Middle Childhood – Main Characteristics

- Age 6-12 years (or 5-12 years)
- Formal schooling years through to adolescence
 - Experience of middle childhood different in developing countries
- Key developmental tasks:
 - Develop concrete operational thought (Piaget)
 - Industry - competence (Erikson)
 - Strengthen self and peer relationships

3

PHYSICAL DEVELOPMENT

- Growth is slow and steady - less rapid than in earlier years; period of calm before growth spurt of adolescence
- 6-7cm per year
- 2-3kg per year



4

Changes in Physical Development

- Average height/weight for boys and girls:
 - 6 year old 115cm/22kg;
 - 12 year old 148cm/47kg
- Varies widely according to ethnic group
- Has altered over history



5

Changes in Physical Development

- Bones continue to grow and harden
- Muscle mass increases as 'baby fat' decreases
- Muscles strengthen
- Early puberty for some children

6

Changes in Physical Development: Brain

- Brain becomes further organised to support mental and physical behaviours in which child is engaged
 - ▣ Further synaptic pruning (increase in connections in areas used; decrease of connections in areas not used)
 - ▣ Focused activation vs diffuse activation in some areas
 - ▣ Advances in functioning of pre-frontal cortex

7

Gross Motor Skills

- Running speed increases
- Can skip and sideways step; fluidity increases
- Ability to throw, kick, dribble, and bat ball improves (body parts coordinate better; increased speed, distance, and accuracy)
- Ability to catch ball improves
- Can jump higher
- Can ride bicycle, skate, climb, skip rope

8

Gross Motor Skills

Gross motor skill improvement

- Agility (ability to move easily and quickly)
- Balance
- Flexibility
- Force

Gender differences:

- Boys have greater force (due to greater muscle mass)
 - Girls have better balance & agility (e.g., hopping, skipping)
- Practice and social expectations may play a role.

9

Fine Motor Skills

- Writing tends to be large at first but becomes smaller over time; legibility increases
- Drawing shows increased organisation, detail, & representation of depth; objects depicted in considerable detail & related to one another as part of organised whole (part cognitive)
- Can fasten buttons, tie shoelaces, build models, execute movements in craft-work
- Improvements due to myelination of central nervous system

10

Fine Motor Skills: Example Art



11

Fine Motor Skills

Gender differences?

- Girls are ahead in tasks that require manual dexterity/fine-motor skills - handwriting and drawing
- Practice and social expectations may play a role

12

Health & Wellness: Illness & Accidents

- Generally a healthy period of life
- Cancer, though uncommon, is leading medical cause of death
- Accidents can lead to death and injury:
 - Death (motor vehicle accident, drowning, fire/burns)
 - Injury (falling, struck by something, unintentional cut)
- Maori children face health disparities

13

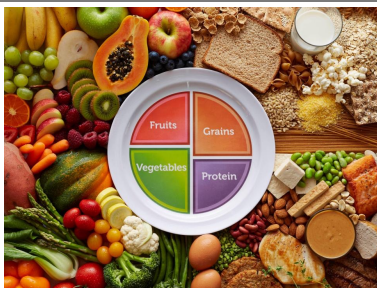
Health & Wellness: Nutrition

- 3 healthy meals – with 2-3 healthy snacks in between
- Smaller portion size than adults
- Keep meals fun; eat as a family
- Stop eating when full
- Encourage trying of new foods
- Takeaways less than once a week
- Plenty of drinks/water
- Limit fruit juice & dried fruit (sugary)
- Offer many different foods from different food groups



14

Health & Wellness: Nutrition



15

Discussion activity

60 min moderate to vigorous exercise per day is recommended and often concern is raised that children of this age don't get enough exercise.

In groups discuss:

- what might impact physical activity at this age group
- positive and negatives
- what limits or encourages physical activity
- why may this age child not have 60 minutes physical activity each day
- how can we enhance their chance to get 60 minutes per day?

16

Health & Wellness: Physical Activity

- Physical activity: 60 minutes moderate/vigorous exercise per day recommended
- Enhances:
 - ▣ Physical fitness
 - ▣ Cognitive and executive functioning (attention, memory, planning & goal-directed activity) – academic success
 - ▣ Psychological functioning – better self image
- Positively impacted by parents, safe neighbourhoods, and schools offering opportunities for physical activity
- Negatively impacted by TV & computer time, chauffeuring children everywhere

17

Health & Wellness: Obesity

- Obesity: 12.3% of NZ children (aged 2-14)
- Increases risk of medical & psychological problems
 - ▣ Overweight children more likely to develop diabetes
 - ▣ Overweight children teased more by peers & family
 - ▣ Obesity linked to low self-esteem in children
- However, important to be aware of anti-fat discourses

18

Discussion activity

Groups to discuss:

- What factors might contribute to obesity?
- What might have changed in the last decades?
- Speed activity – 2 minutes

Feedback

19

Health & Wellness: Obesity

- Hereditary and environment contribute to obesity
 - ▣ Food environment/practices have changed in last decades
 - More convenience/processed food in market
 - Nutritious food expensive
 - Changes in labour force (more working mothers and two-income families means less time to prepare healthy, nutritious meals)
 - Larger portion sizes
 - ▣ More screen time for children (at expense of exercise)
 - ▣ Adversity, insecurity, stress, and emotional turmoil (eventually leading to junk food self-medication)

20

Health & Wellness: Obesity

- In developed world, children in most deprived neighbourhoods have higher obesity than those in least deprived neighbourhoods
 - ▣ Healthy lifestyles less accessible
 - ▣ Less access to parks and playgrounds (fewer of them; lower perceived safety) & extracurricular activities
 - ▣ Decreased availability of nutritious food (expensive; more takeaway outlets in deprived neighbourhoods)
 - ▣ Greater adversity, stress, and emotional turmoil
 - ▣ Greater susceptibility to marketing?

21

How can child obesity be addressed?

- Ideas?
- Reducing poverty, increasing education, and providing support for families (upstream drivers)
- Combined with more balanced nutrition, reduced junk food marketing, and increased physical activity traditional downstream drivers (downstream drivers)

22

Health & Wellness: Sleep

- Sleep is very important for children's wellbeing
- 9-11 hours per night required



23

COGNITIVE DEVELOPMENT

- Children make enormous strides in their cognitive development during middle childhood as their thought processes and language become more logical and complex.

24

Piaget: Concrete-operational stage

Piaget: The Concrete-Operational Stage

Piaget's Periods of Cognitive Development			
Birth to 2 years	Sensori-motor	Uses senses and motor skills, items known by use	Object permanence learned
2-6 yrs	Pre-operational	Symbolic thinking, language used; egocentric thinking	Imagination/experience grow, child de-centers
7-11 yrs	Concrete operational	Logic applied, has objective/rational interpretations	Conservation, numbers, ideas, classifications
12 yrs to adulthood	Formal operational	Thinks abstractly, hypothetical ideas (broader issues)	Ethics, politics, social/moral issues explored

25

Piaget: Concrete-operational stage

Piaget: The Concrete-Operational Stage

- Emerges around 7 years and continues to 11
- Can reason logically as long as applied to specific or concrete examples
- Concrete-operational thought is reversible (able to mentally go through a series of steps in a problem and then reverse the direction, returning to the starting point)
- Children can reverse mathematical operations (e.g., $2 + 3 = 5$ can be reversed to $5 - 3 = 2$)
- Children less egocentric (and more socio-centric)
- Able to engage in decentration (focus on multiple parts of a problem at once)
- Understand laws of conservation, seriation, transitivity

26

Piaget: Conservation

- Understands law of conservation

Piaget's Conservation Tasks			
Conservation Task	Age of Acquisition	Original Presentation	Transformation
Number	6-7 years		
Liquid	6-7 years		
Mass	6-7 years		
Area	8-10 years		

27

Piaget: Conservation

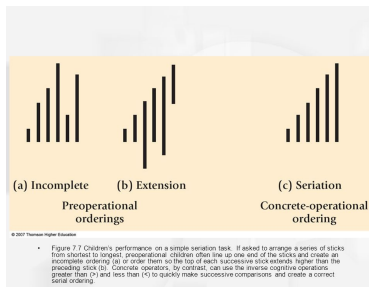
- Video about conservation at different ages



28

Piaget: Seriation

- Can place objects in a series by height, weight, age

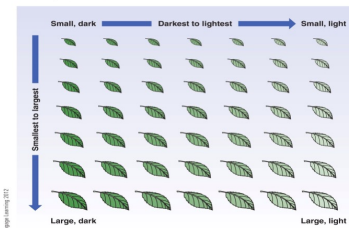


29

Piaget: Seriation

- Children can seriate two dimensions at once

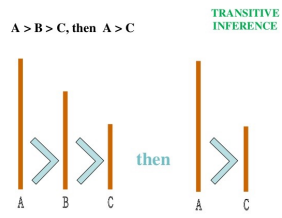
To classify these leaves, children must focus on two dimensions at once: size and lightness. They must also understand the principle of transitivity—that if $A > B$ and $B > C$, then $A > C$.



30

Piaget: Transitivity

- If A exceeds B in some property and B exceeds C, then A must also exceed C



31

Beyond Piaget

Some of Piaget's conclusions have been challenged:

- Different concrete operational abilities do not emerge at same time
- Education and culture can influence children's development
 - ▣ Some pre-operational children can be trained to reason at concrete operational stage
 - ▣ Development of conservation skills depends on practice within a culture

32

Information processing approach

- Alternative to Piaget's theory of cognitive development
- Emphasises how children use:
 - ▣ Attention
 - ▣ Memory
 - ▣ Strategies to process information

33

Language development

- Children gain new skills at school that make reading and writing possible
 - ▣ Language to talk about things that are not physically present
 - ▣ Learning what a word is
 - ▣ Learning how to recognise and talk about sounds
 - ▣ Learn the letters of the alphabet represent sounds of language

34

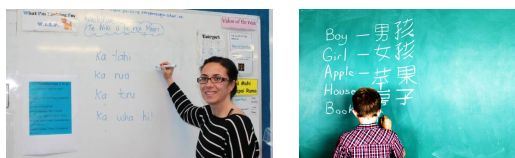
Language development

- Children's vocabulary increases from about 14,000 words at 6 years to 40,000 words by 11 years
- Children's grammar advances

35

Language development

- Bilingualism has a positive effect on children's cognitive development



36

Inclusive Education

Inclusive education is:

- embedded in inclusive values
- is a social justice and human rights issue
- sees all students as capable learners
- supports all learners by adapting the environment, intentionally removing barriers to learning so that all students can be present, participate, learn and belong

(Booth & Ainscow, 2016; Carrington & MacArthur, 2012; Ministry of Education, n.d.-a)

37

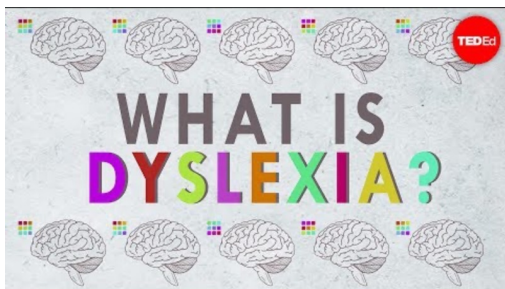
Learning differences

- A child with a learning difference may struggle with listening, thinking, speaking, reading, hand-writing, written language, spelling and/or maths.
- Specific learning disabilities:
 - Dyslexia (reading and spelling)
 - Dysgraphia (handwriting)
 - Dyscalculia (maths)
 - Dyspraxia (motor function)

38

Learning differences: Dyslexia

- Video about dyslexia



39

Learning differences: Dyslexia

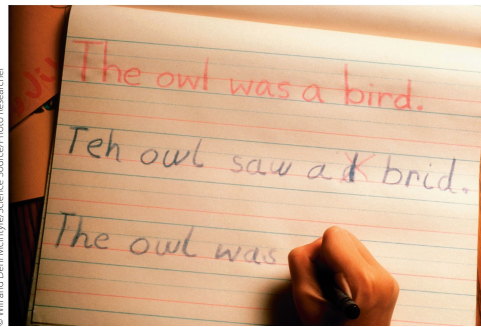
Important to identify dyslexia early

Signs (that persist after one year of schooling):

- unsure of sound a letter/combination makes, e.g. h, sh
- poor sense of rhyme
- often mixes up letters, e.g., b/d/p/q
- or mixes numbers, e.g., 15 for 51
- finds learning sequences difficult, e.g., months
- very poor reading, even of very familiar words
- great difficulty spelling simple words like 'duck' or 'like'
- highly disorganised – often doesn't know what day it is

40

Dyslexic children may perceive letters as upside down (confusing w with m) or reversed (confusing b with d), leading to rotations or reversals in writing, as shown here.



41

ADHD

Inattentive

Easily distracted
Short attention span
Day dreams
Makes silly mistakes
Often late
Dis-organised

Hyperactive

Always moving
Restless
Fidgets
Can't sit still
Trouble switching off / sleeping

Impulsive

Acts without thinking
Interrupts people
Accident prone
Blurts out answers / secrets

Types:

- Predominantly inattentive ADHD
- Predominantly hyperactive/impulsive ADHD
- Combined ADHD
- Not a learning disability – but ADHD can affect learning

42

Medical model vs social model of disability

- "Fix" the child vs 'fix' the environment
- Deficit view and inclusive views on disability
- Attitude can be a barrier
- Label to enable



Wikimedia Commons

43

SOCIAL-EMOTIONAL DEVELOPMENT

- Development of Self-Understanding
- Understanding Others
- Self-esteem & Self-concept
- Self-efficacy
- Self-regulation
- Industry vs Inferiority
- Emotional Development
- Moral Development & Pro-Social Behaviour
- Peer relationships

44

Development of self-understanding

- Children in middle childhood increasingly describe themselves in terms of psychological characteristics and traits
 - "I amnice... helpful... mean... smart... dumb... funny... naughty... shy..." etc.
- Describe themselves in terms of social groups
 - "I am on the soccer team."
- Compare what they can do with what others can do
 - "I am the best speller in the class."

45

Understanding others

- Describe others in terms of psychological traits and social groups
 - E.g., *"Kate is a very good reader and is also good at art and science. She's nice to everyone in our class and she's funny. She takes piano lessons and likes to play soccer."*

46

Understanding others

- Perspective taking improves – better able to consider the perspective of others and understanding their thoughts and feeling
 - Involves cognitive inhibition: controlling their own thoughts to consider the perspective of others
 - Involves cognitive flexibility: seeing situations in different ways
- Improved perspective-taking allows children to get on better with their peers.

47

Self-esteem & self-concept

- **Self-esteem** is the *global* evaluation of self, e.g., a good kid or a naughty kid
 - Perception does not always match reality, i.e., inflated or overly-negative
- **Self-concept** is a *domain-specific* evaluation of self, e.g., academic, athletic, attractiveness, social
 - Some domains may matter more than others; not averaged
- Self-esteem drops from early to middle childhood due to comparisons with others
- Affected by quality of parent-child interactions (secure, affectionate, & involved vs abusive/neglectful) and peer interactions (positive vs bullied)
- Low-self esteem linked to current/later anxiety, depression, suicide

48

Self-efficacy (Bandura)

- Self-efficacy is the belief one can master a situation and produce favourable outcomes



- Helplessness is the belief one cannot master a situation or produce favourable outcomes

49

Self-efficacy (Bandura)

- Children with high self-efficacy expect they will be able to do well at an activity/learning task
- More likely to be enthusiastic, apply effort, and persist longer at a learning task
- Children with low self-efficacy/helplessness believe they cannot do well - and avoid or give up on task
- Important parents and teachers do not contribute to low self-efficacy

50

Self-regulation

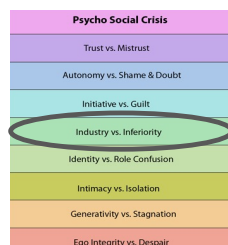
- Increased capacity to regulate one's behaviour, emotions, and thoughts
- Leads to increased social competence and achievement
- Linked to development advances in brain's pre-frontal cortex



51

Industry vs Inferiority (Erikson)

- Industry vs inferiority is fourth stage of eight in Erikson's stages of psychosocial development (occurs 8-11 years)



52

Industry vs Inferiority

- Psychosocial Conflict:** Industry versus Inferiority
- Major Question:** "How can I be good?"
- Basic Virtue:** Competence
- Important Event(s):** School
- School and social interaction play an important role during middle childhood

53

Industry vs Inferiority

- Some children may discover that their abilities are better than those of their friends or that their talents are highly prized by others
 - Competence & industry, pride & confidence
- Some children discover that they are less capable than other children and/have little or no encouragement from parents, teachers, or peers
 - Doubt, inferiority

54

Emotional development

- Increased ability to talk about one's own and others' emotions - & to control emotions to meet social standards

55

Emotional development

- Improved emotional understanding, e.g., shame, pride
- Increased capacity for genuine empathy
- Increased tendency to be aware of events leading to emotional reaction
- Ability to suppress or conceal negative emotional reactions, e.g., to control anger when irritated
- Increasingly able to use strategies for controlling emotions, e.g., soothing oneself after an upset

56

Moral development (Piaget)

- From age 5-7 years, children are in a stage of **moral realism**
 - They believe rules are created by wise adults, must be followed, and cannot be changed; believe breaking a rule leads to punishment
- From age 8, children progress to stage of **moral relativism**
 - They believe rules are created by people to help them get along; they can be changed if necessary
 - Emerges due to advances in cognitive development (able to understand reasons for rules) & interactions with peers

57

Moral development

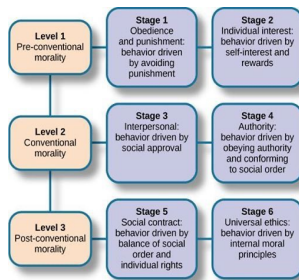
- Video on Kohlberg's theory of moral development



58

Moral development (Kohlberg)

- Level 1: Pre-conventional level



59

Moral development (Kohlberg)

Level 1: Pre-conventional level

- Stage 1: **Obedience orientation**
 - Children obey because adults tell them to obey. Children base their moral decisions on fear of punishment
- Stage 2: **Individual interest / instrumental orientation**
 - Children look out for their own interests – and expect any favours to be returned.
 - E.g., if you are nice to others, others will be nice to you

60

Prosocial behaviour: Helping others

- Intentions to act prosocially increases with age – as do children's strategies for helping
- Some children behave more prosocially than others; some more likely to look out for own interests

61

Prosocial behaviour: Helping others

Skills underlying prosocial behaviour

- Perspective taking – the better children understand thoughts and feelings of other people, more willing to share and help
- Empathy – children who deeply feel another person's fear, disappointment, sadness, or loneliness are more inclined to help
- Moral reasoning – more likely to help if there is a chance of reward (or avoidance of punishment) or if child thinks behaviour will be reciprocated

62

Gender differences in social-emotional development

- Boys more physically aggressive than girls (occurs in all cultures)
- Less self-regulation of emotion and lower self-control can translate to behavioural problems
 - ▣ In NZ, boys more likely to have problematic scores on Strengths & Difficulties questionnaire (SDQ)
 - ▣ SDQ is a behavioural screening questionnaire; measures emotional symptoms, conduct problems, hyperactivity/inattention, peer relationship problems, prosocial behaviour
- Girls engage in more prosocial behaviour
 - ▣ View themselves as more empathetic & share slightly more
 - ▣ More kind and considerate behaviour

63

Peer relationships

- Peer relationships especially important in middle childhood
- Linked to positive outcomes
- Until age 12, children prefer same-sex friendships

64

Peer relationships

- Five types of peer status
 - ▣ **Popular children** (frequently nominated as a best friend and are rarely disliked by their peers - have number of social skills)
 - ▣ **Average children** (receive an average number of both positive and negative nomination from peers)
 - ▣ **Neglected children** (infrequently nominated as a best friend, but are not disliked by their peers)
 - ▣ **Rejected children** (infrequently nominated as a best friend and are actively disliked by their peers - often have serious adjustment problems)
 - ▣ **Controversial children** (frequently nominated as best friend and as being disliked by peers)

65

Bullying

Definition

- **Bullying is deliberate** - harming another person intentionally
- **Bullying involves a misuse of power in a relationship**
- **Bullying is usually not a one-off** - it is repeated, or has the potential to be repeated over time
- **Bullying involves behaviour that can cause harm** - it is not a normal part of growing up.
- Bullying is when one student (or a group of students) keeps picking on another student again and again to make them feel bad. They say or do things to upset them, make fun of them, stop them from joining in, or keep hitting or punching them.

66

Bullying

- Relative common in middle childhood
- Can be overt (easily observed) or covert (indirect or hidden)
- Rarely occurs in front of adults

67

Bullying: Roles

- Initiators
 - Students who bully others often do so to gain status and recognition from their peers. Reinforced when peer group colludes instead of challenges behaviour
- Targets
 - Unassertive/withdrawn
 - Different in some way from majority of students at school
 - Have disability
 - New at school
- Bystanders/observers
 - Have a powerful effect on encouraging or inhibiting bullying behaviour

68

Bullying

- Bystanders
 - followers (assistants) – do not initiate, but take an active role in the bullying behaviour
 - supporters (reinforcers) – support the bullying behaviour (overtly or covertly, e.g., by turning a blind eye) but do not take an active role in the bullying behaviour
 - defenders – dislike the bullying and try to help the target by intervening, getting teacher support (using safe telling) or providing direct support to the target.

69

Bullying

- Whole school approach
 - Real change happens when students, staff, parents and whānau, and other members of the community share responsibility for making their school a respectful and inclusive environment.
- Focus on developing healthy social behaviours and strategies (rather than only addressing bullying behaviours)
- Teacher supervision in playground
- Confidential reporting system

70

Bullying

- Programmes
 - Positive Behaviour for Learning (PB4L)
 - Kia Kaha
 - Wellbeing@school
 - KiVa

71

PROTECTIVE & RISK FACTORS

- Middle childhood is a critical time in development when the advantages or disadvantages of early childhood can be maintained, reversed, or advanced.
- So, what factors are protective and what factors increase risk

72

Risk factors

- Poverty
- Family violence
- Having special education needs (a learning disability or an emotional behavioural problem)
- Bullying

73

Protective factors

- Positive parent-child/whanau relationships
- Presence of friendships
- Cultural identity (e.g., for Māori)

74

WHAT CONTRIBUTES TO WELLBEING IN MIDDLE CHILDHOOD?

- Key dimensions of wellbeing from perspective of children in middle childhood:
 - ▣ **Quality relationships** with parents, friends, wider family, pets, teachers (loving, caring, accepting, listening, supportive, togetherness & belonging, fair)
 - ▣ **Quality environments** – home, school, and neighbourhood
 - ▣ **Playing, having fun, enjoying activities** at home and across neighbourhood environments
 - ▣ **Special events** (holidays, celebrations like birthdays and Christmas)
 - ▣ **Security & safety** (physiological needs met & absence of bullying)
 - ▣ **Being healthy**
 - ▣ **Accomplishment** – including sports accomplishment

75

Features of middle childhood

- What are the key features of physical, cognitive, and social-emotional development in middle childhood?



76
